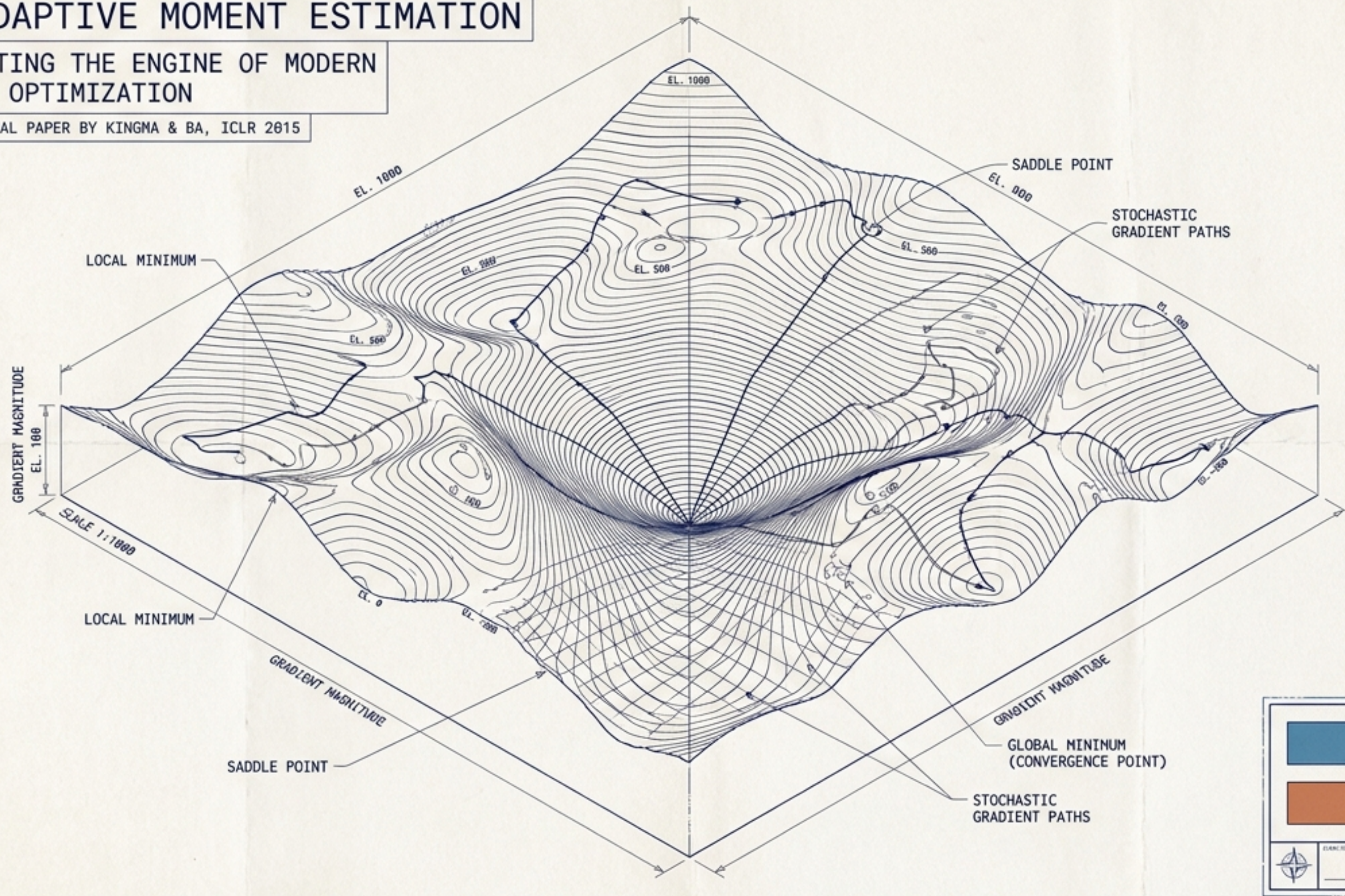


DECONSTRUCTING THE ENGINE OF MODERN STOCHASTIC OPTIMIZATION

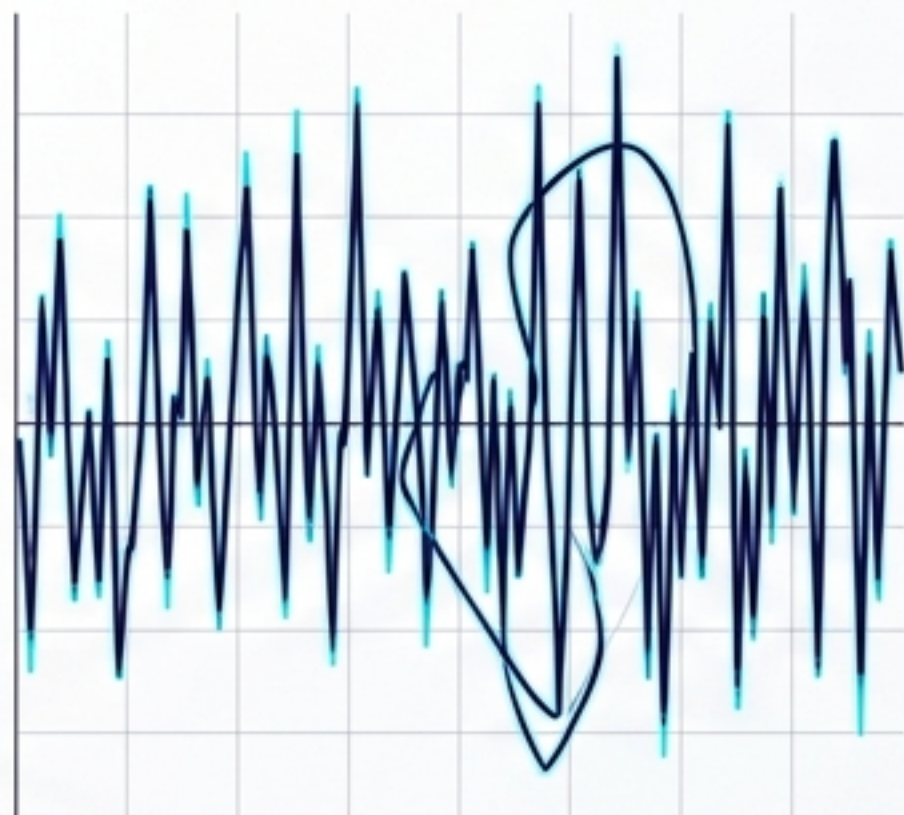
BASED ON THE SEMINAL PAPER BY KINGMA & BA, ICLR 2015



PROJECT: NEURAL NETWORK OPTIMIZATION

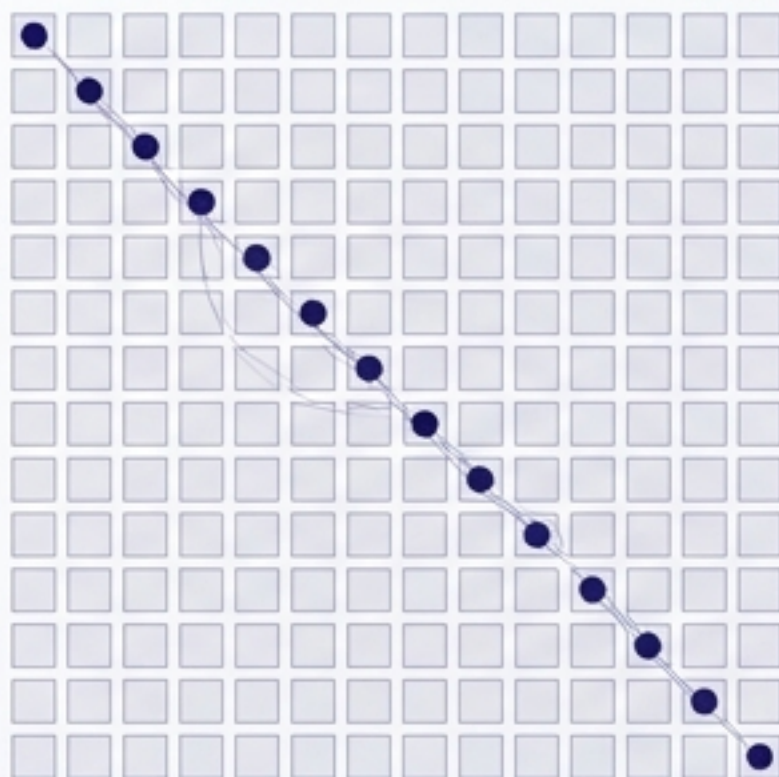
The physical limits of first-order optimization

NOISE



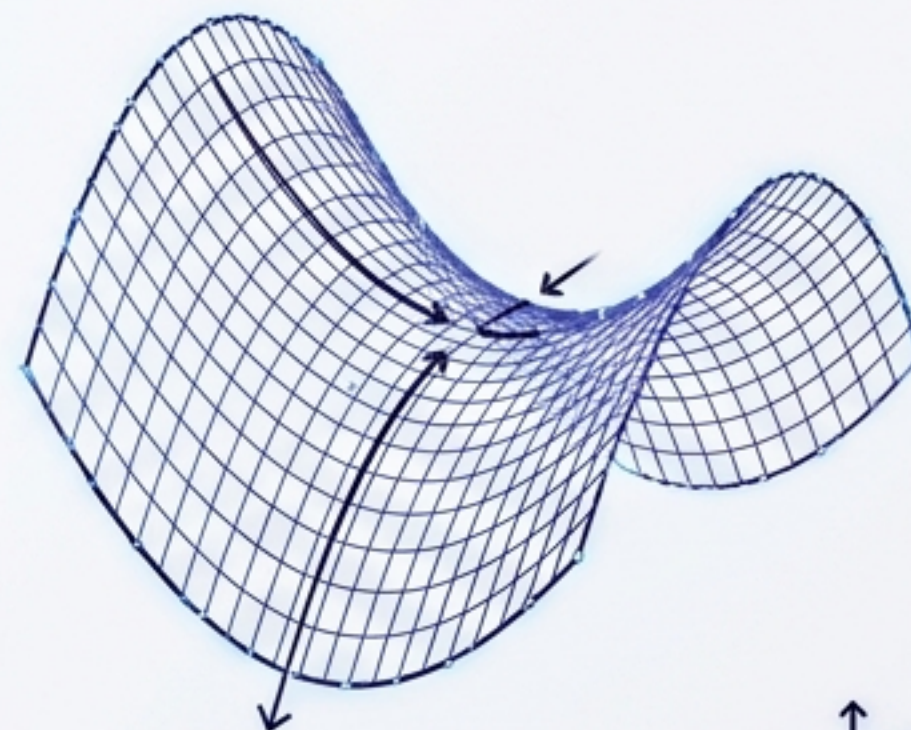
Objective functions are highly stochastic or non-stationary, causing standard gradient descent to oscillate wildly.

SPARSITY



Features appear infrequently. Standard optimizers apply uniform learning rates, failing to aggressively learn rare but critical features.

HIGH DIMENSIONALITY



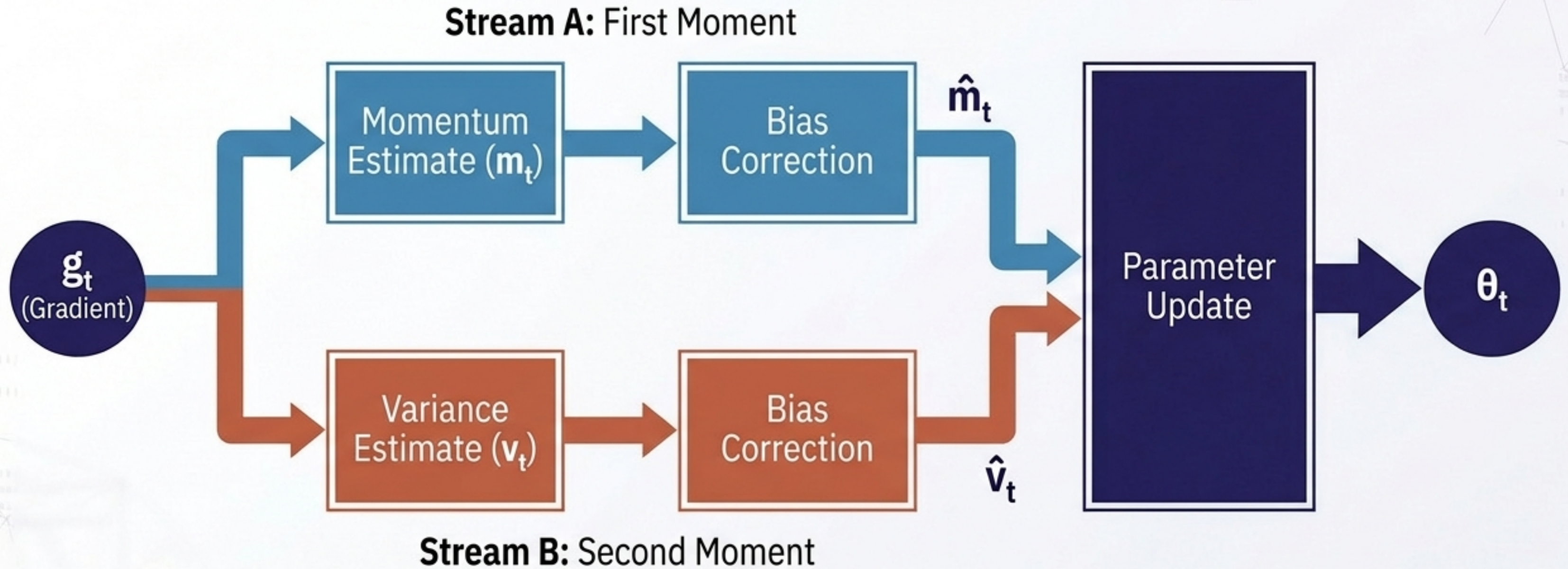
Complex parameter spaces are filled with saddle points where higher-order methods become computationally infeasible.

The evolutionary synthesis of stochastic optimization

Adam synthesizes AdaGrad's sparsity handling with RMSProp's non-stationary resilience, while introducing a crucial missing mathematical component: bias correction.

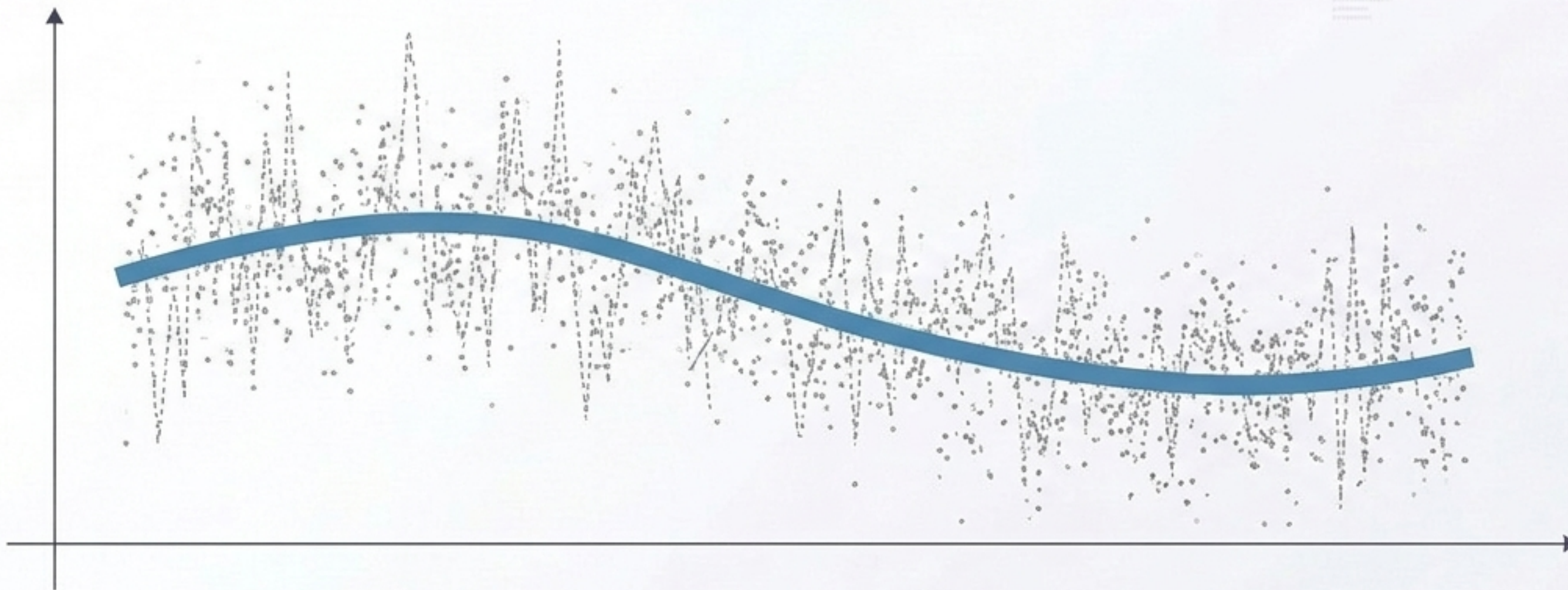
	Vanilla SGD	AdaGrad	RMSProp	Adam
Scales with Data/Params	●	●	●	●
Adapts to Sparse Gradients	×	●	×	●
Thrives in Non-Stationary Settings	×	×	●	●
Initialization Bias Corrected	×	×	×	●

The Adam architecture pipeline



Adam computes **individual adaptive learning rates** for different parameters from estimates of **first** and **second** moments of the gradients.

Stream A computes gradient momentum



$$\mathbf{m}_t = \beta_1 \cdot \mathbf{m}_{t-1} + (1 - \beta_1) \cdot \mathbf{g}_t$$

Biased first moment estimate

$$\mathbf{m}_t = \beta_1 \cdot \mathbf{m}_{t-1} + (1 - \beta_1) \cdot \mathbf{g}_t$$

Exponential decay rate (default 0.9)

This stream estimates the mean of the gradient. It acts like a heavy ball rolling down a hill, accumulating momentum in the right direction and ignoring local noise.

Stream B scales for terrain variance



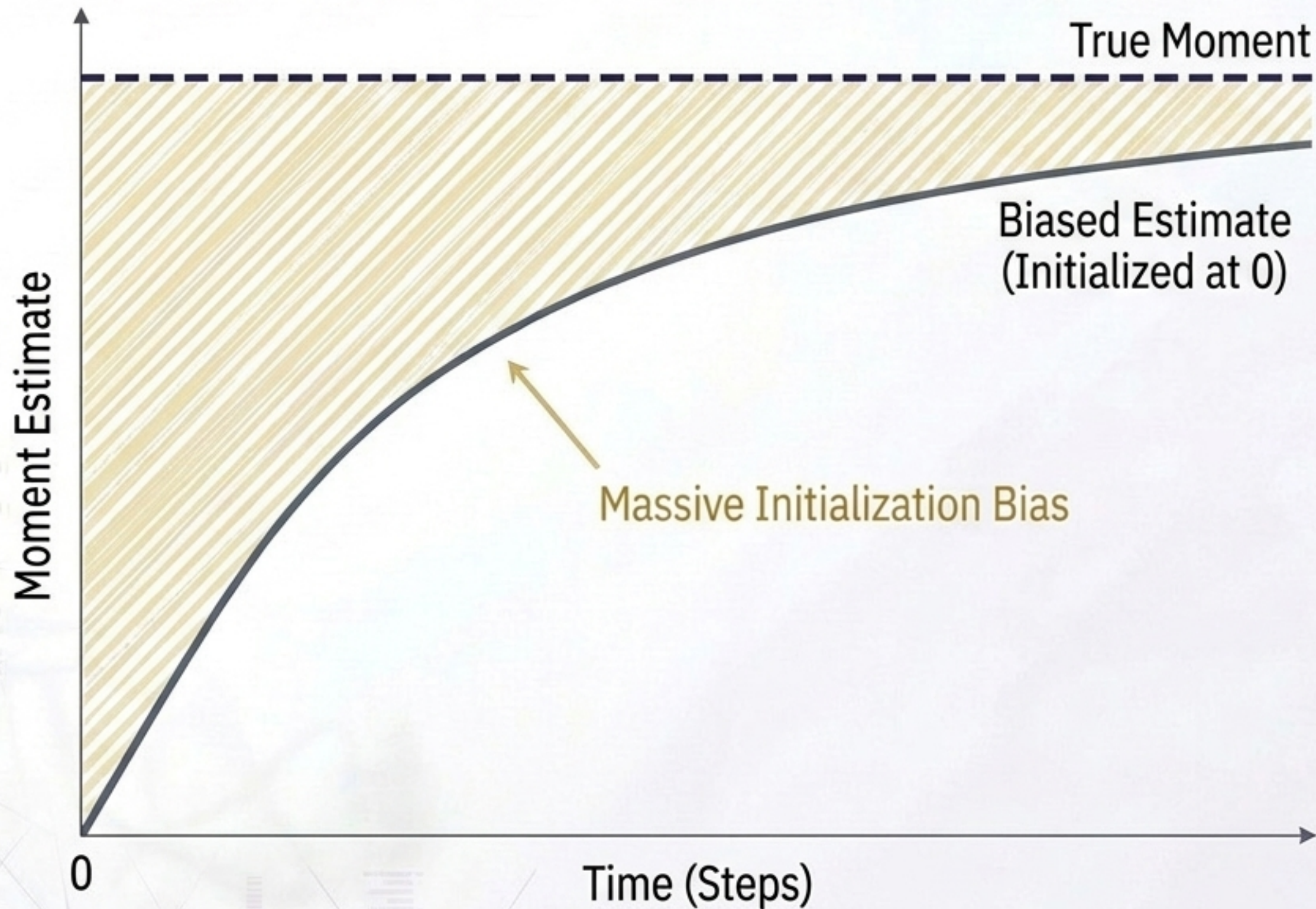
$$\mathbf{v}_t = \beta_2 \cdot \mathbf{v}_{t-1} + (1 - \beta_2) \cdot \mathbf{g}_t^2$$

This stream estimates the uncentered variance. It acts as a dampener, automatically reducing the step size for parameters that experience massive, frequent updates and boosting the step size for sparse parameters.

Decay rate
(default 0.999)

Biased second raw moment estimate

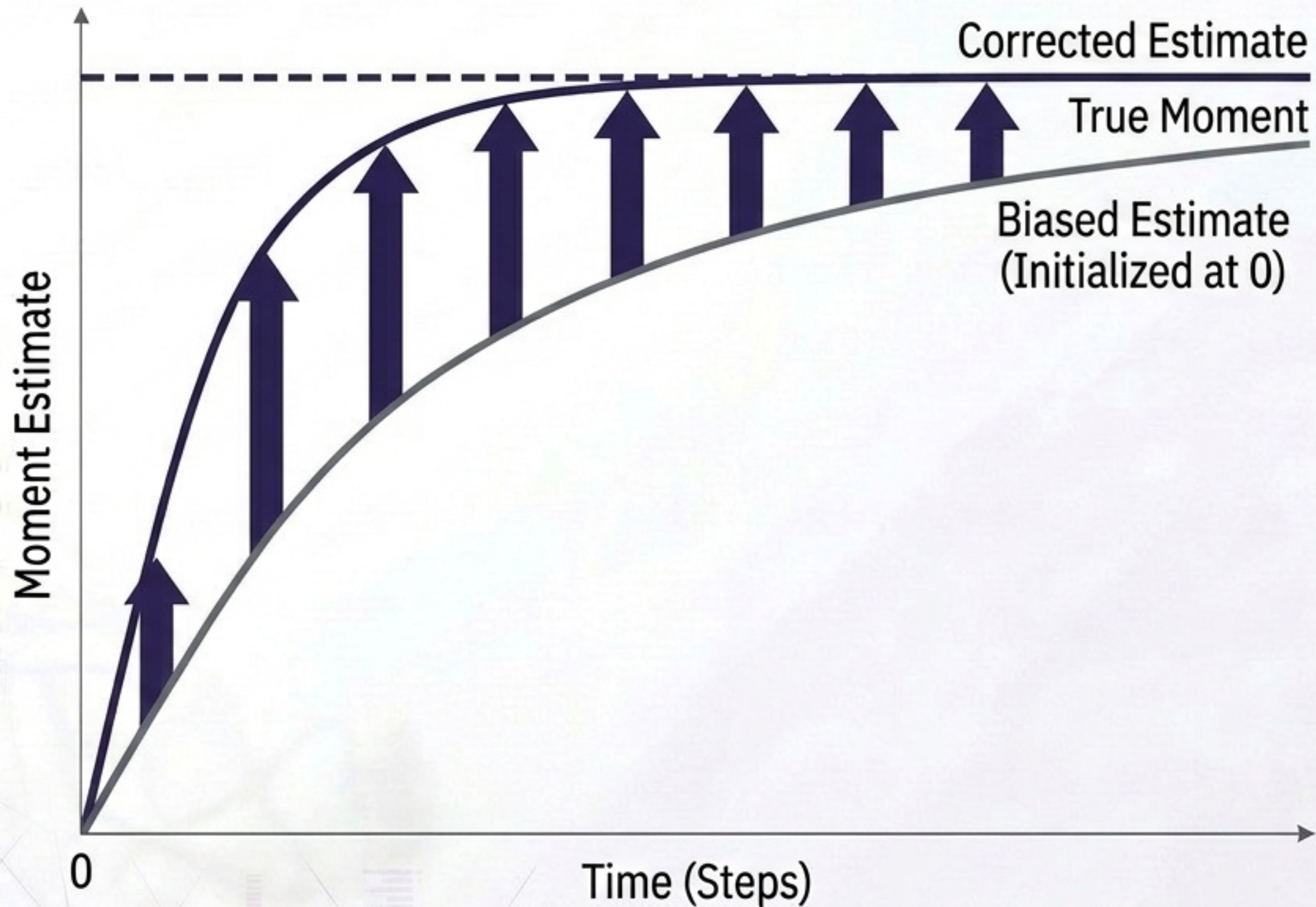
The initialization bias trap



Because moving averages are initialized as vectors of zeros, early estimates are heavily biased toward zero.

This is especially dangerous during initial timesteps when decay rates are close to 1, dragging estimates down and crippling the optimizer's cold start.

Mathematical counterweights restore true moments

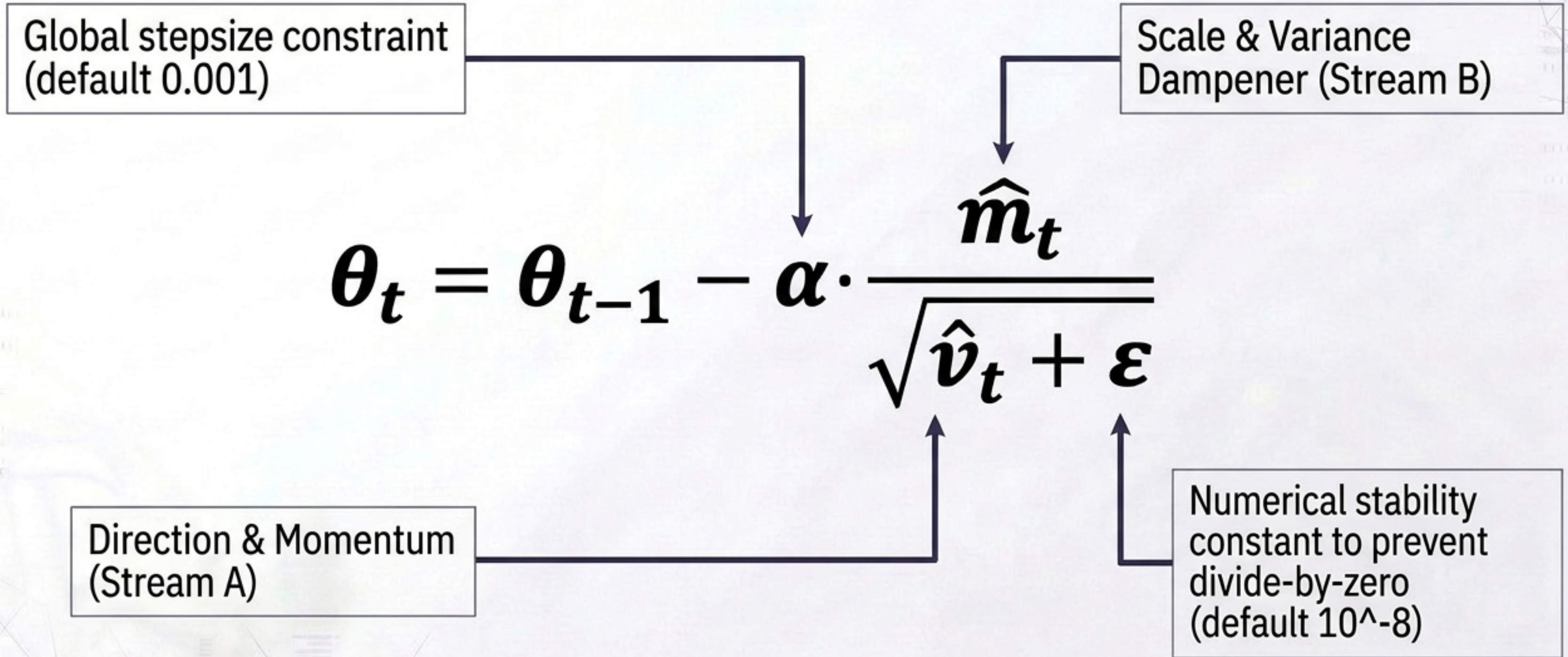


$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$$

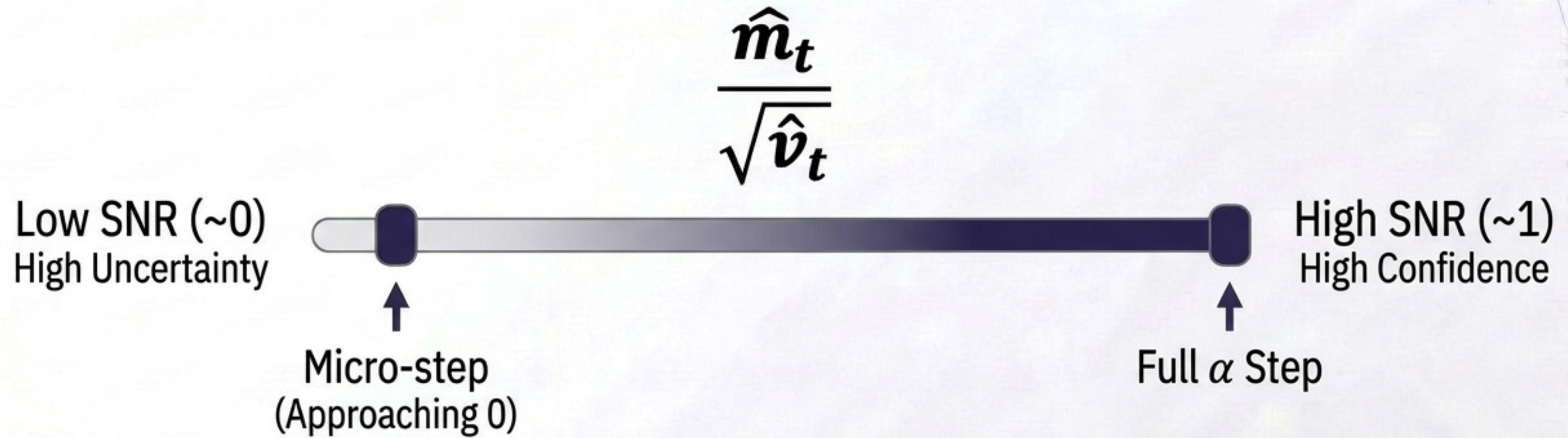
$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

By dividing by the decay rate to the power of t , Adam instantly corrects the initialization bias. The t exponent ensures this correction is powerful at step 1, but safely vanishes to a divisor of 1 as $t \rightarrow \infty$.

The assembled parameter update

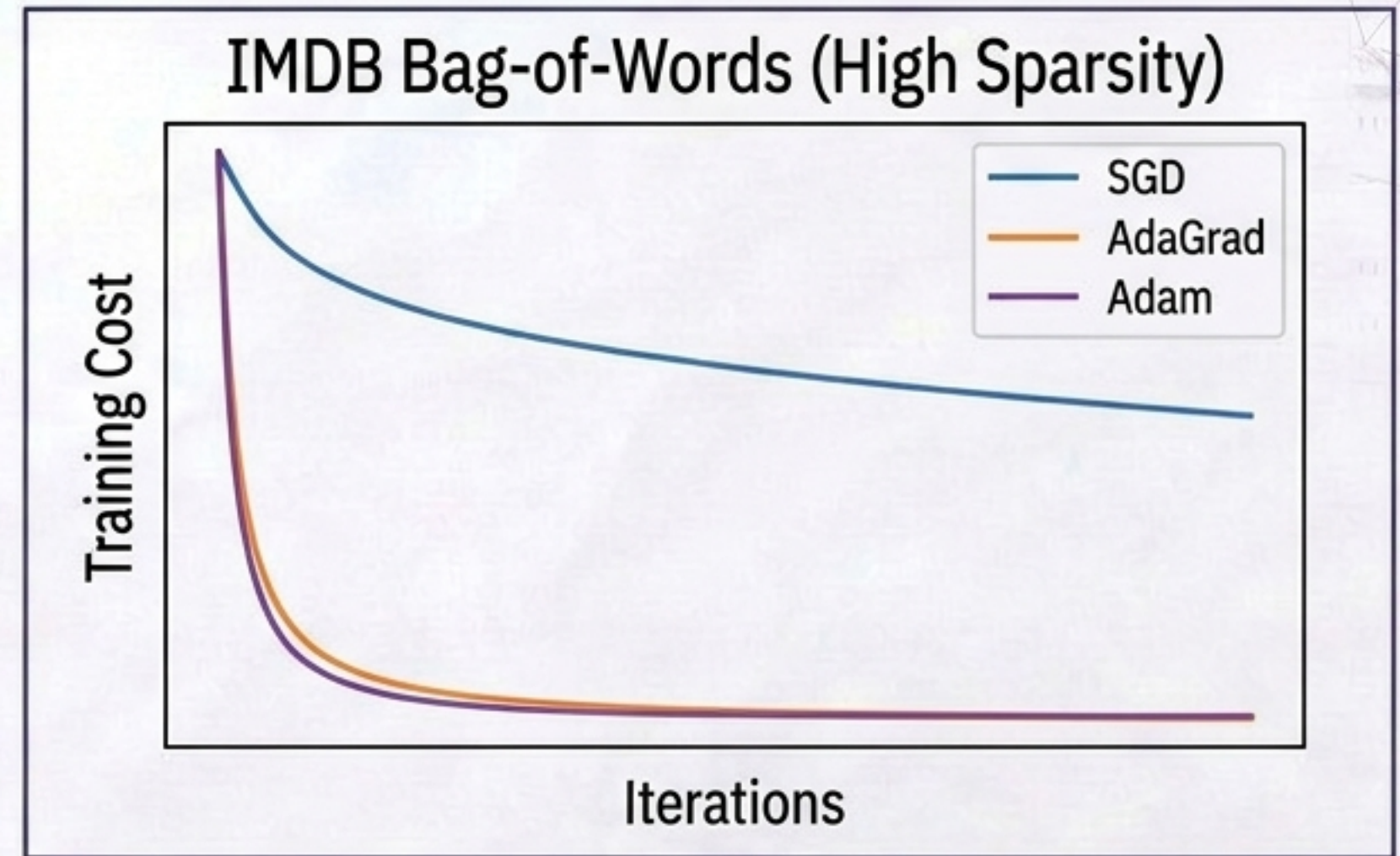
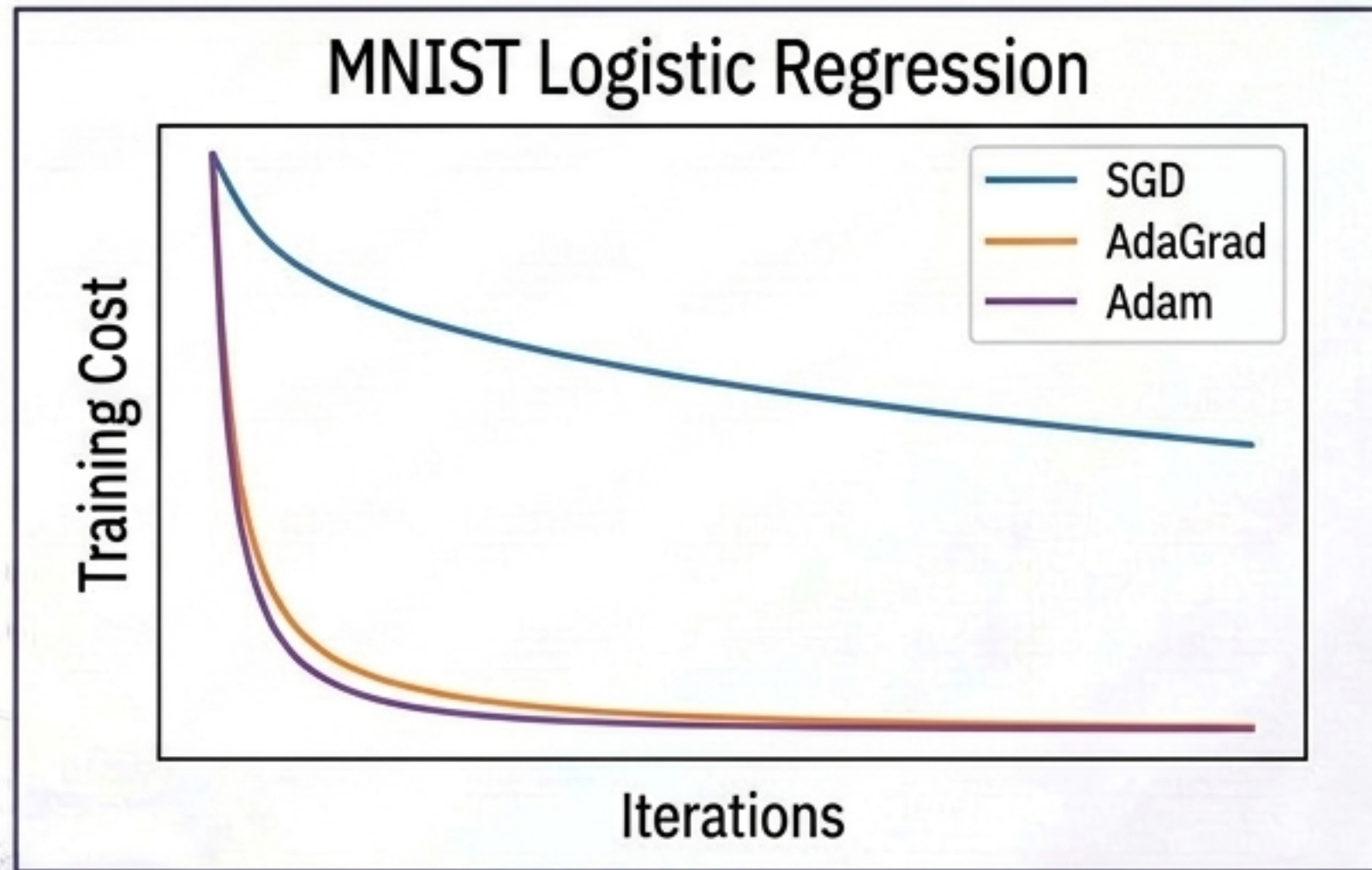


Automatic annealing via signal-to-noise



Adam establishes a 'trust region' around current parameters. The maximum effective step size is strictly bounded by α . When the signal is clear, Adam takes a full step. When uncertainty increases (SNR approaches 0 near an optimum), the step size automatically anneals toward zero without manual intervention.

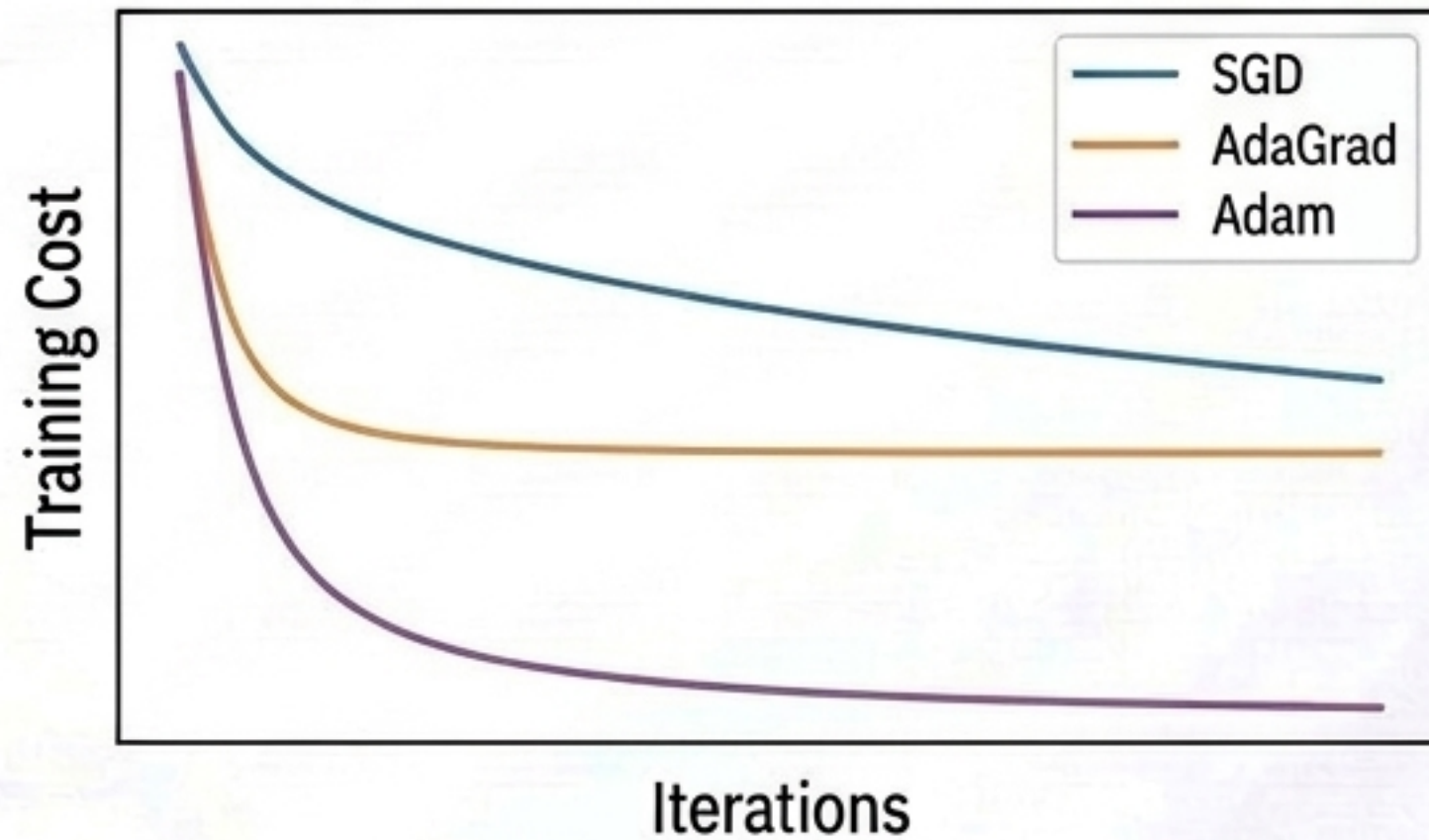
Matching AdaGrad on sparse feature spaces



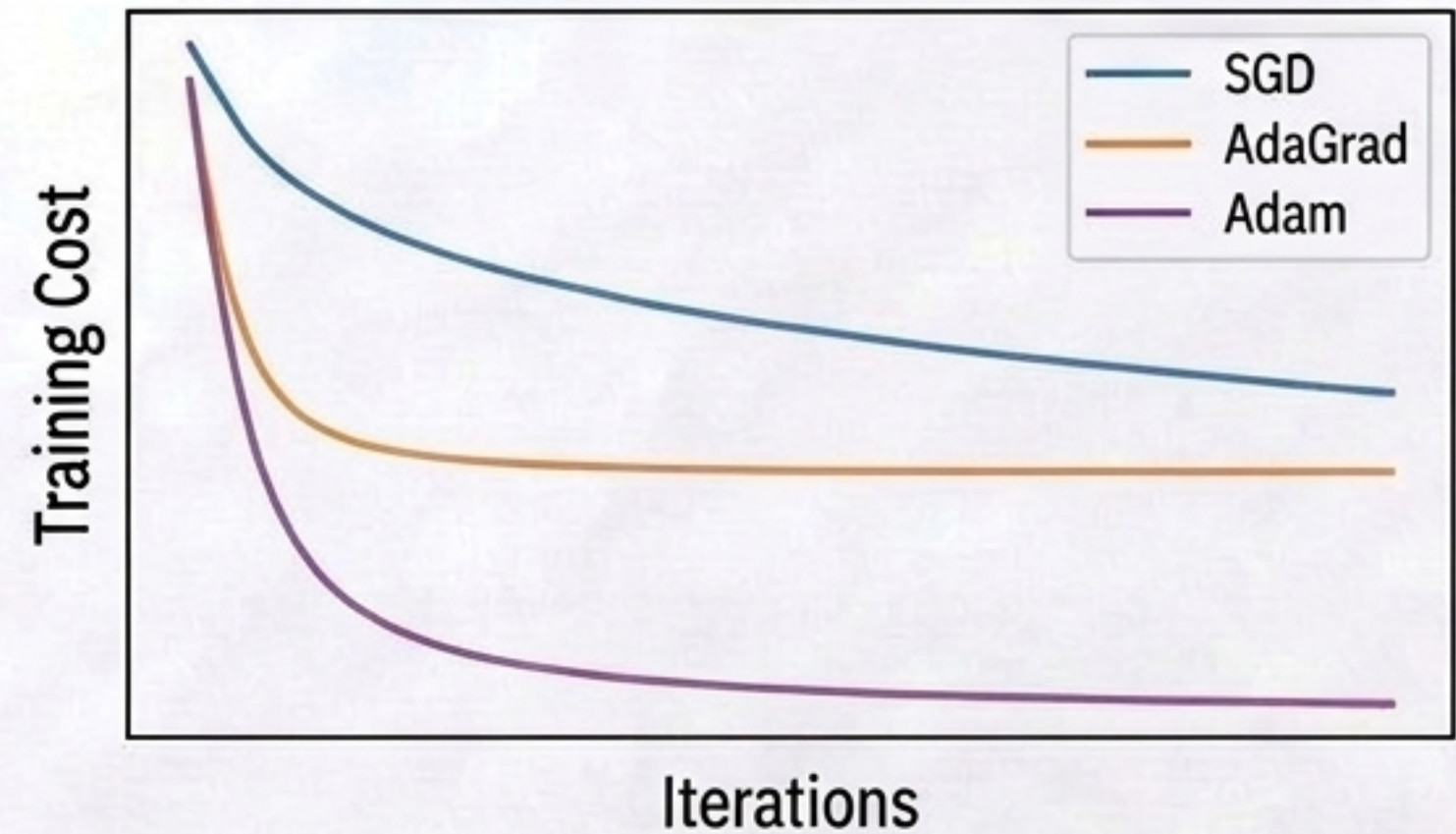
On highly sparse features, standard SGD fails to learn rare data points quickly. AdaGrad thrives here by design. Adam automatically matches AdaGrad's rapid convergence rate while requiring far less manual tuning.

Dominating deep and noisy landscapes

Multi-layer Neural Network with Dropout

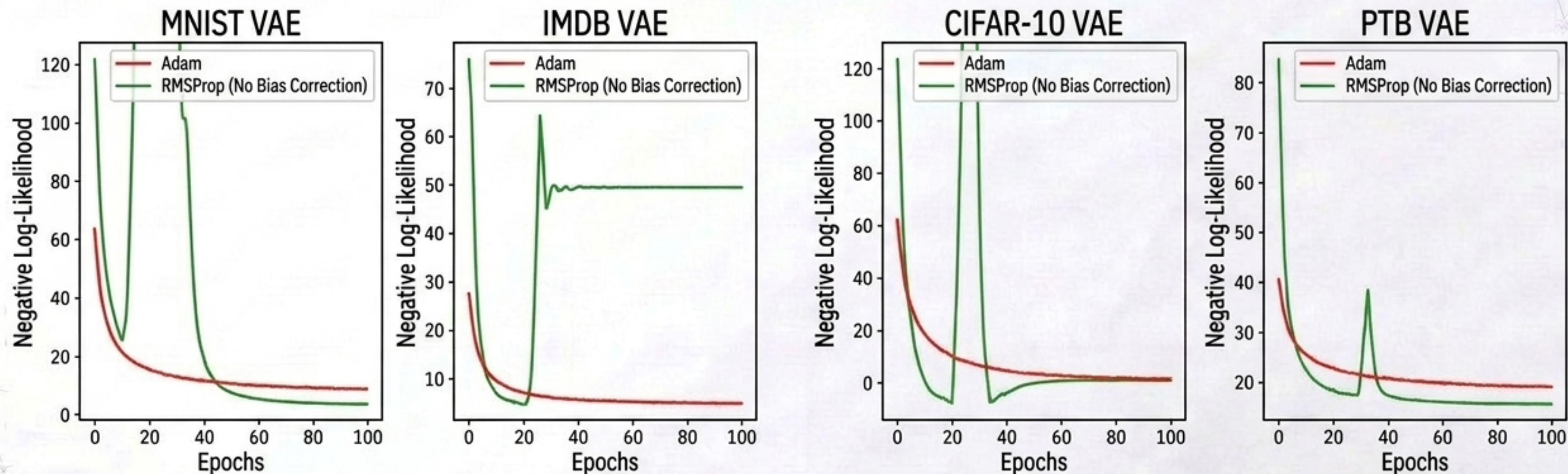


Deep CNN (CIFAR10)



In deep architectures, gradients vary vastly across layers. AdaGrad's denominator accumulates infinitely, permanently stalling learning. Adam's moving average forgets the distant past, allowing it to adapt layer-by-layer and conquer stochastic noise.

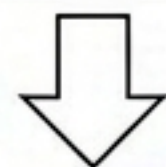
The physical danger of omitting bias correction



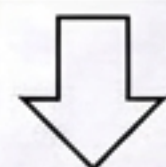
Training a VAE with sparse gradients requires a β_2 close to 1. Without bias correction (RMSProp - shown in green), this causes a massive initial bias, leading to infinitely large, unstable parameter updates and catastrophic divergence. Adam's correction (shown in red) ensures absolute stability across all hyperparameter settings.

AdaMax: Generalizing to the infinity norm

L_p norm update rule where $p \rightarrow \infty$



$$u_t = \lim_{p \rightarrow \infty} (v_t)^{1/p}$$



$$u_t = \max(\beta_2 \cdot u_{t-1}, |g_t|)$$

Scaling the update rule to an L_∞ norm yields a remarkably simple, completely stable recursive variant. AdaMax provides a simpler absolute bound on parameter updates and, conveniently, requires zero initialization bias correction.

The default engine of modern deep learning

Computational Efficiency

Requires minimal memory overhead (only maintaining two moment vectors) and scales effortlessly to massive datasets and high-dimensional parameter spaces.

Unrivaled Versatility

Perfectly combines the specialized sparsity expertise of AdaGrad with the resilient non-stationary capabilities of RMSProp into a single algorithm.

Intuitive Defaults

The out-of-the-box hyperparameters ($\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) work exceptionally reliably across almost every architecture, cementing its legacy.